

 Edit



Implement AWS Analytics Services by Using Amazon Athena

Challenge Overview

Understand the scenario

You are a cloud Administrator for Hexelo, an organization that needs to perform queries against data stored in S3 buckets. You have decided to use Amazon Athena to perform the queries. First, you will create S3 buckets to store data and query results. Next, you will configure Amazon Athena. Finally, you will perform queries against the data using Amazon Athena.

Navigating the Challenge Lab

▼ Understanding a Cloud Slice

Your Challenge Lab is using a cloud slice that reflects a dynamic, constantly evolving cloud environment. A cloud slice is a subset of a cloud platform subscription that has been assigned to a user account which was provisioned for you to use for the duration of this Challenge Lab. It provides temporary access to a subset of resources available in a cloud subscription so that you can learn the concepts without having to configure your own subscription. A cloud slice has restrictions on the types of administrative activities that are allowed.

As you complete your Challenge Lab, you may find that the provided guidance is not identical to what you encounter in your cloud slice environment. If you encounter a difference between your cloud slice and the Challenge Lab instructions, please let us know by submitting feedback directly to [Challenge Labs feedback](#) so that we may update the content in as timely a manner as possible.

▼ Quick tips for navigating the Challenge Lab instructions.

✎ Select the Copy to Clipboard icon to copy the green text.

✎ Select the Type Text icon to insert the green text directly into the Challenge Lab environment.

⚠ An Alert tells you that a task requires extra care.

ℹ A Note provides additional helpful information for completing a task.

💡 A Hint will guide you through a portion of the Challenge Lab.

📖 A Knowledge block provides a deeper level of knowledge into a subject. It is a great way to solidify your understanding, but it is not strictly necessary to complete the Challenge Lab.

Create prerequisite resources

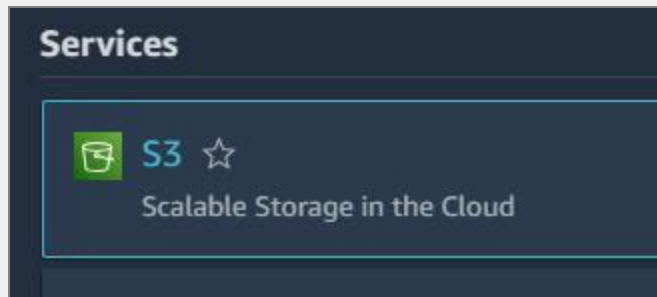
Hints Enabled

No ☒ Yes

- Using the S3 console, create an S3 bucket named `data-bucket-{LAB_INSTANCE_ID}`.

💡 Expand this hint for guidance on creating a bucket.

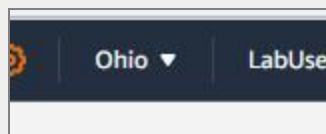
- In the upper-left corner of the page, select **AWS** to return to the AWS Management Console.
- In the AWS Management Console, in Search for services, search for, and then select  **S3**.



- Select **Create bucket**.



- In AWS Region, ensure that **US East (Ohio)** is selected.



- In Bucket name, enter  **data-bucket-{LAB_INSTANCE_ID}**.

Note: The bucket name must be globally unique.

- Review the default settings, and then select **Create bucket** to create the bucket.

💡 Want to learn more? Review the documentation on [creating a bucket](#).

- Create a second S3 bucket named  **results-bucket-{LAB_INSTANCE_ID}**.

 You should now be able to see your buckets listed.

- Download the file below to your local hard drive as  [moviedata.csv](#).

[moviedata.csv](#)

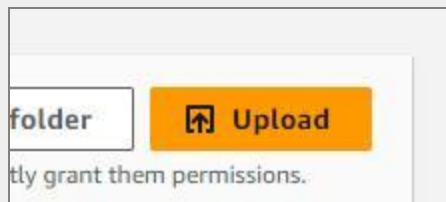
 Expand this hint for guidance on downloading a file. 

- Right-click the [moviedata.csv](#), and then select **Save link as**.
- Download the file below to your local hard drive as  [moviedata.csv](#).

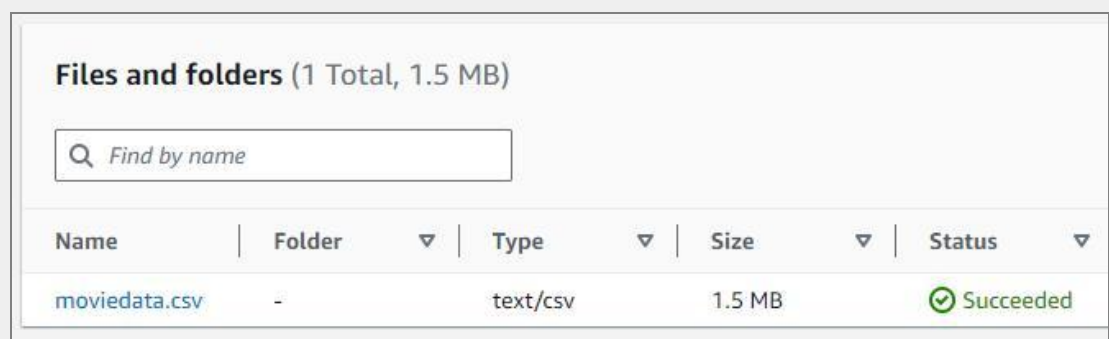
- Upload the **moviedata.csv** file from you local drive to the  [data-bucket-{LAB_INSTANCE_ID}](#) bucket.

 Expand this hint for guidance on uploading a file. 

- On the Amazon S3|Buckets page, in General purpose buckets, select the **data-bucket-{LAB_INSTANCE_ID}** bucket (the link, not the radio button).
- On the **data-bucket-{LAB_INSTANCE_ID}** bucket page, in Objects, select **Upload**.



- On the Upload page, in Files and folders, select **Add files**.
- Select the **moviedata.csv** file, and then select **Open**.
- Select **Upload**.





Want to learn more? Review the documentation on [uploading a file to a bucket](#).

Check your work

Verify

Configure Amazon Athena

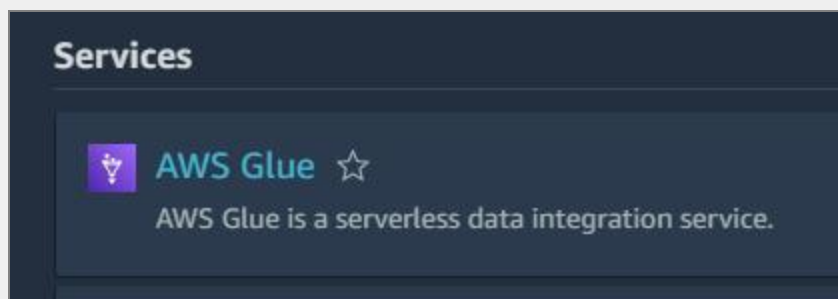
Hints Enabled

No ☒ Yes

- In the AWS Glue console, create an AWS Glue database named  **moviedata**.

💡 Expand this hint for guidance on an AWS Glue database.

- In the upper-left corner of the page, select **AWS** to return to the AWS Management Console.
- In the AWS Management Console, in Search for services, search for, and then select  **AWS Glue**.



- On the AWS Glue navigation menu, expand Data Catalog, and then select **Databases**.



- On the Databases page, select **Add database**.



- On the Create a database page, in Database details, in Name, enter  **moviedata**, and then select **Create database**.

Create a database

Create a database in the AWS Glue Data Catalog.

Database details

Name

Database name is required, in lowercase characters, and no longer than 255 characters.

Location - optional

Set the URI location for use by clients of the Data Catalog.

Description - optional

Enter text

Descriptions can be up to 2048 characters long.

Cancel Create database

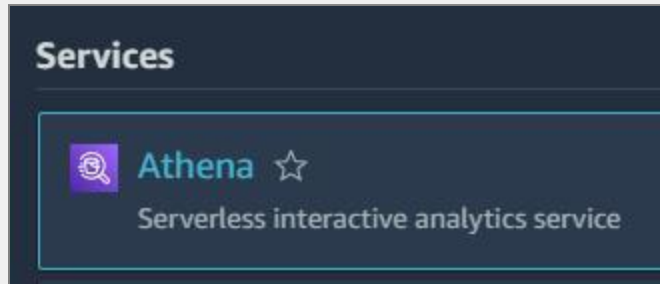


Want to learn more? Review the documentation on [creating an AWS Glue database](#).

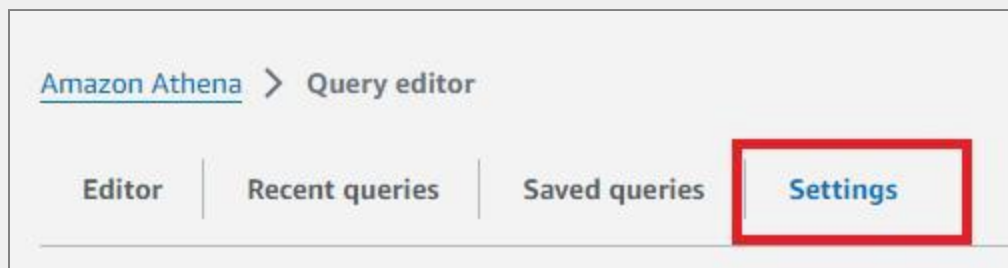
- Use the Amazon Athena console to configure the query result location as **results-bucket-{LAB_INSTANCE_ID}**.

💡 Expand this hint for guidance on configuring a query result in the Amazon Athena console. ^

- In the upper-left corner of the page, select **AWS** to return to the AWS Management Console.
- In the AWS Management Console, in Search for services, search for, and then select  **Athena**.



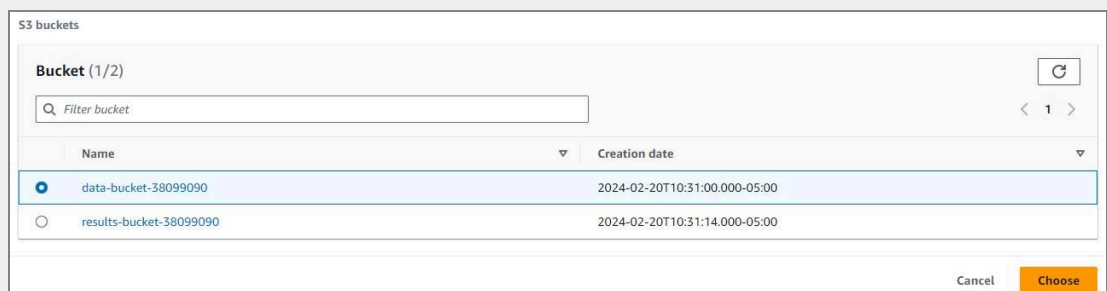
- On the Query editor page, select the **Settings** tab.



- On the Settings tab, select **Manage**.



- On the Manage settings page, in Query result location and encryption, select **Browse* S3**.
- In Choose S3 data set, in S3 buckets, select **results-bucket-{LAB_INSTANCE_ID}**, and then select **Choose**.



- On the Manage settings page, select **Save**.

Amazon Athena > Query editor > Manage settings

Manage settings

Query result location and encryption

Location of query result - optional
Enter an S3 prefix in the current region where the query result will be saved as an object.

Q s3://data-bucket-38099090 X View Browse S3

You can create and manage lifecycle rules for this bucket
Use Amazon S3 lifecycle rules to store your query results and metadata cost effectively or to delete them after a period of time.
[Learn more](#)

Lifecycle configuration

Expected bucket owner - optional
Specify the AWS account ID that you expect to be the owner of your query results output location bucket.

Enter AWS account ID

☐ Assign bucket owner full control over query results
Enabling this option grants the owner of the S3 query results bucket full control over the query results. This means that if your query result location is owned by another account, you grant full control over your query results to the other account.

☐ Encrypt query results

Cancel Save

 Want to learn more? Review the documentation on [configuring a query result in the Amazon Athena console](#).

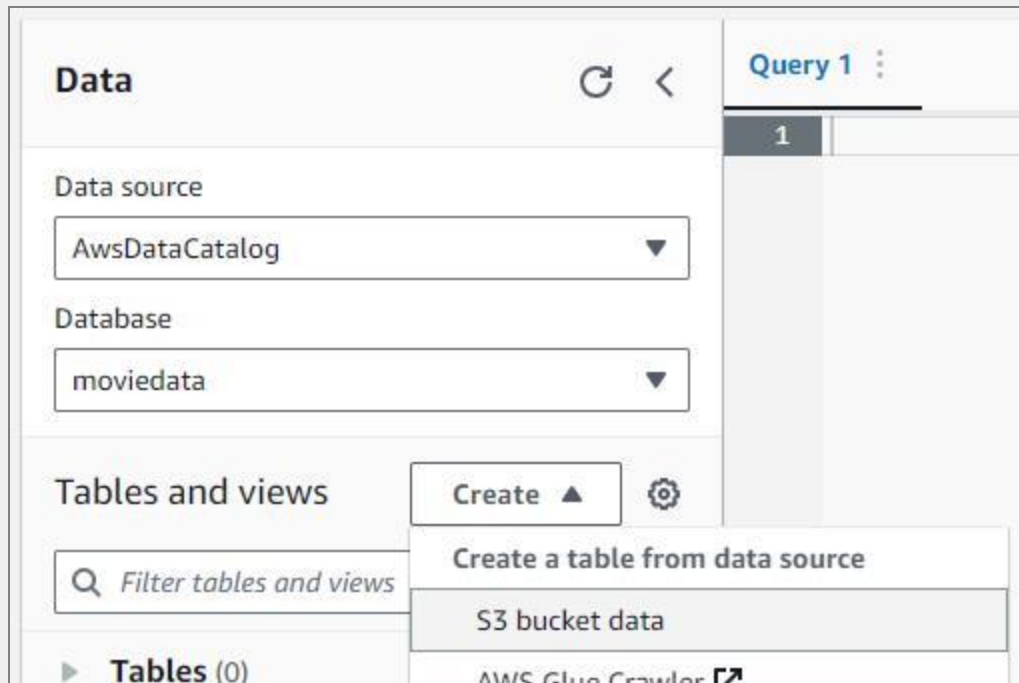
- In Amazon Athena, begin creating a new table from **S3 bucket data** by using the following details:

Property	Value
Table Name	 movietable
Database	moviedata
Dataset	 data-bucket-{LAB_INSTANCE_ID}
File format	CSV
SerDe library	OpenCSVSerde

Property	Value
separatorChar	 ,
quoteChar	 "
escapeChar	 \

💡 Expand this hint for guidance on creating a new table from S3 bucket data.

- In the AWS Management Console, select Amazon Athena to return to the Query editor.
- On the Editor tab, in the Data section, select **Create**, and then select **S3 bucket data**.



- On the Create table from S3 bucket data page, in Table details, in Table name, enter **movietable**.
- In database configuration, select Choose an existing database, and then select **moviedata**.

Table details

Table name

Table name must be from 1-128 characters and must be unique. Valid characters are a-z, A-Z, 0-9, _(underscore). Table names tend to correspond to the directory where the data will be stored.

Description - optional

Table description must be from 1-1024 characters. 1024 characters remaining.

Database configuration [Info](#)

Choose an existing database or create a new database

Choose to access an existing database or to create a new database in order to create a new table. Athena stores the table schema in the AWS Glue Data Catalog.

☐ Create a database

☒ Choose an existing database

- In Dataset, select Browse S3, select **data-bucket-{LAB_INSTANCE_ID}**, and then select **Choose**.

Choose S3 data set

S3 buckets

Bucket (1/2)

< 1 >

Name	Creation date
☒ data-bucket-38099090	2024-02-20T10:31:00.000-05:00
☐ results-bucket-38099090	2024-02-20T10:31:14.000-05:00

Cancel Choose

- In Data format, in File format, select **CSV**, and then in in SerDe library, select **OpenCSVSerde**.

Data format [Info](#)

Table type

Apache Hive

File format

CSV

SerDe library

org.apache.hadoop.hive.serde2.OpenCSVSerde

- In SerDe properties, in Name, in separatorChar, enter a value of , in quoteChar, enter a value of ", and then in escapeChar, enter a value of \\.

SerDe properties - optional

Name

Value

separatorChar

,

Remove

quoteChar

"

Remove

escapeChar

\\

Remove

Add SerDe property

- Do not create the table yet.



Want to learn more? Review the documentation on [creating a new table from S3 bucket data](#).

- In **Column details**, add the following columns:

Column name	Column type
 title	string
 year	string
 plot	string
 rating	string

Column name	Column type
time	string

💡 Expand this hint for guidance on adding column details to a table.



- In Column details, In Column name, enter **title**, in Column type, select **String**, and then select **Add a column**.
- In Column details, In Column name, enter **year**, in Column type, select **String**, and then select **Add a column**.
- In Column details, In Column name, enter **plot**, in Column type, select **String**, and then select **Add a column**.
- In Column details, In Column name, enter **rating**, in Column type, select **String**, and then select **Add a column**.
- In Column details, In Column name, enter **time**, and then in Column type, select **String**.

Column details

Column name must be from 1-128 characters. Valid characters are a-z, A-Z, 0-9, _(underscore). Certain advanced column types (namely, structs) are not exposed in this interface.

Column name	Column type	Description - optional	
title	string	Enter description	Remove
year	string	Enter description	Remove
plot	string	Enter description	Remove
rating	string	Enter description	Remove
time	string	Enter description	Remove

- Review the remaining default options, and create the table.

💡 Expand this hint for guidance on creating a table

- On the Create table from S3 bucket data, select **Create table**.

Cancel

Create table

Check your work

Verify

Use Amazon Athena to perform queries

Hints Enabled

No ☒ Yes

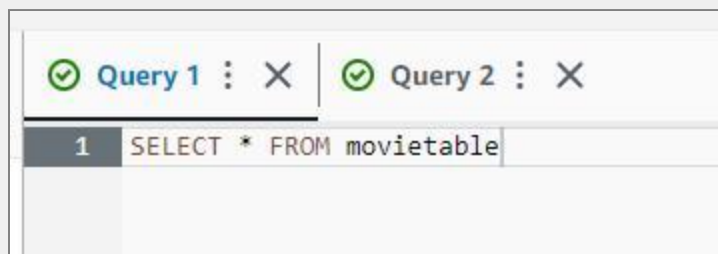
- Use the Amazon Athena console to create a query against [movietable](#) and return all rows.

💡 Expand this hint for guidance on running an Amazon Athena query. ^

Click the + icon to create a new query tab, and then run the following command:

sql

 `SELECT * FROM movietable`



💡 Want to learn more? Review the documentation on [performing a query in Athena](#).

- Perform another query returning only the [title](#), [year](#), [rating](#) and [time](#) columns.

💡 Expand this hint for guidance on running an Amazon Athena query.

Click the + icon to create a new query tab, and then run the following command:

sql

```
SELECT title, year, rating, time
FROM movietable
```

Results (8)					Copy	Download results
Search rows						
< 1 > ⚙️						
# ▾	title ▾	year ▾	rating ▾	time		
1						
2						
3	hive" title* title2 varchar8 @H P "					
4	hive" year* year2 varchar8 @H P "					
5	hive" plot* plot2 varchar8 @H P "+					
6	hive" rating* rating2 varchar8 @H P "					

- Enter the title of the first movie item returned in the textbox below:

❓ Ignore any items that contain **Title** or a blank space.

- Perform another query returning only the  title,  year,  rating and  time columns sorted by  year in descending order.

 Expand this hint for guidance on running an Amazon Athena query.

Click the + icon to create a new query tab, and then run the following command:

sql

```
SELECT title, year, rating, time
FROM movietable
ORDER BY year;
```

- Enter the title of the first item returned in the textbox below:

- Perform another query returning only the  title,  year,  rating and  time columns sorted by  rating in descending order and only display 10 items.

 Expand this hint for guidance on running an Amazon Athena query.

Click the + icon to create a new query tab, and then run the following command:

sql

```
SELECT title, year, rating, time
FROM movietable
ORDER BY rating DESC
limit 10;
```

- Enter the title of the first item returned in the textbox below:

Check your work

Verify

Summary

Congratulations, you have completed the **Implement AWS Analytics Services by Using Amazon Athena** Challenge Lab.

You have accomplished the following:

- Created prerequisite resources
- Configured Amazon Athena
- Used Amazon Athena to perform queries

Your feedback is important!

As you end your Challenge Lab, please take a few minutes to complete the short survey that will appear in the next window.

Alternatively, you may provide your feedback directly to [Challenge Labs feedback](#).

Be sure to check out the other Challenge Labs in this series:

- Monitor AWS access by using IAM tools [Guided]
- Monitor Activity by Using the AWS Audit Manager [Guided]
- Filter Traffic by Using AWS Web Application Firewall (WAF) [Guided]
- Implement Best Practices by Using the AWS Trusted Advisor [Guided]
- Create Resources by Using the Console, CLI and Cloud Formation [Guided]
- Implement Microservices by Using AWS Containers [Guided]
- Manage AWS ML Services by Using Amazon Sagemaker Canvas [Guided]
- Implement AWS Analytics Services by Using Amazon Athena [Guided]
- Create a Message Queue by Using Amazon Simple Queue Service (SQS) [Guided]
- Run Python Code by Using AWS Cloud9 [Guided]
- Implement Cost Management by Using AWS Budgets and AWS Cost Explorer [Guided]
-

